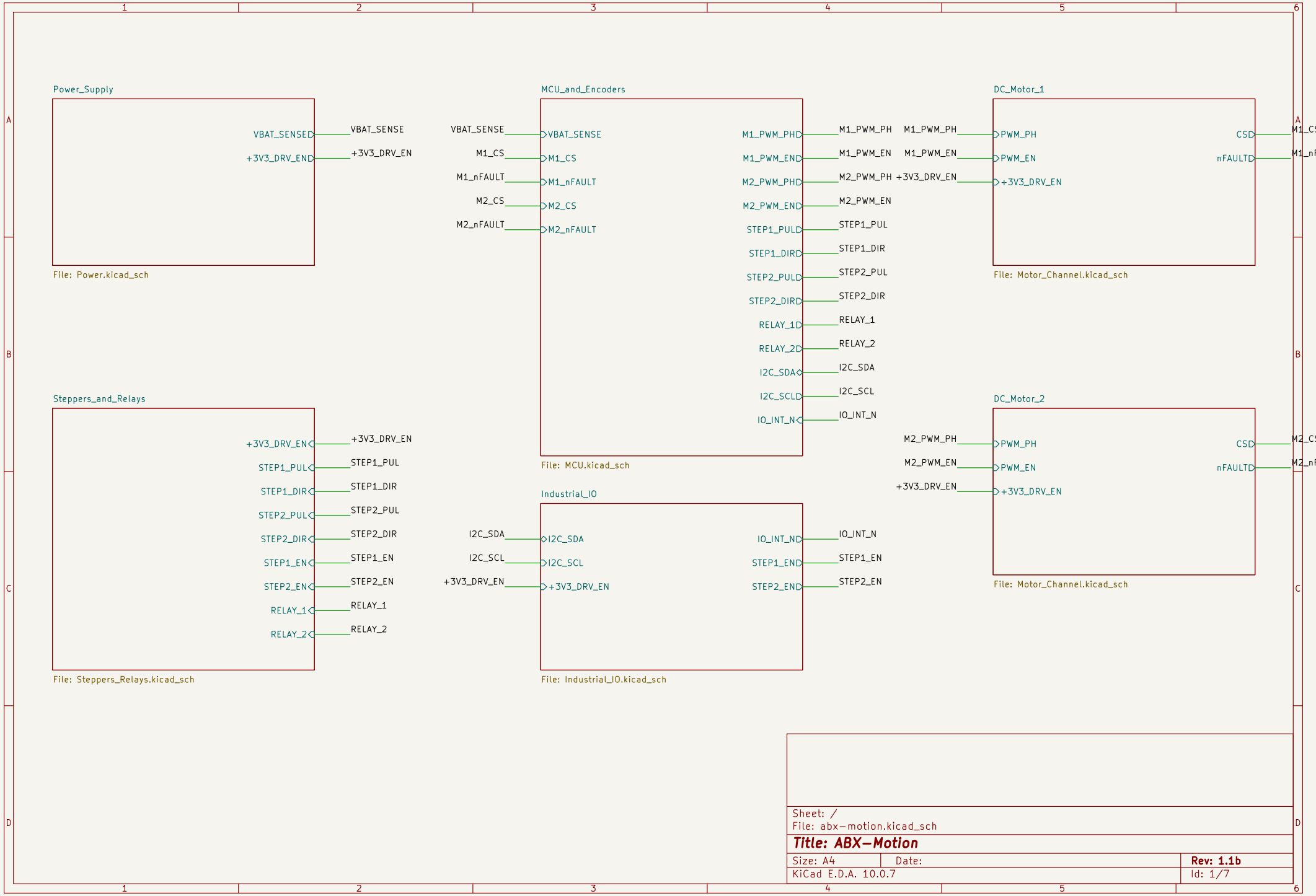
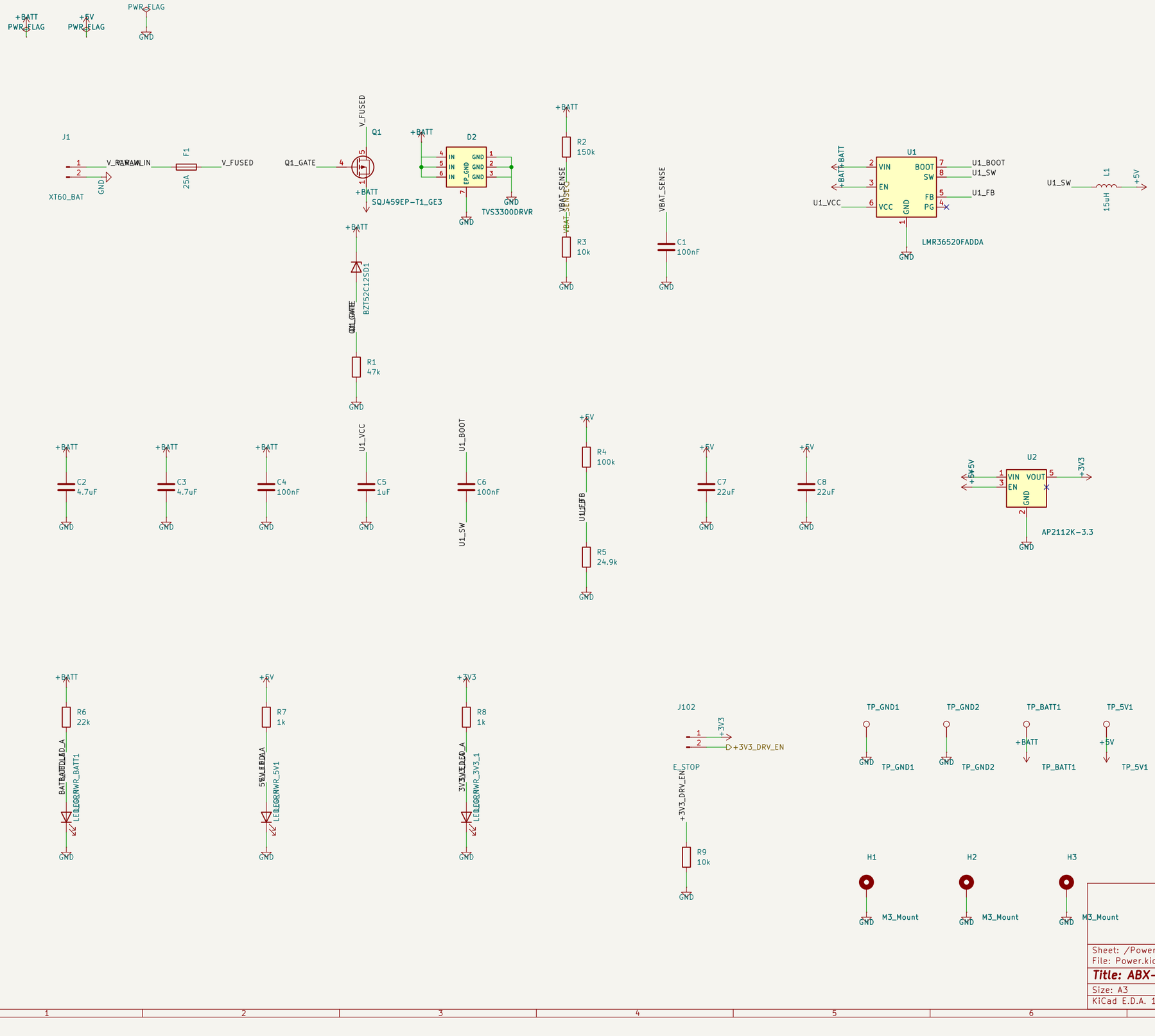

Power_Supply
VBAT_SENSE
+3V3_DRV_EN
VBAT_SENSE
+3V3_DRV_EN
File: Power.kicad_sch
MCU_and_Encoders
VBAT_SENSE
M1_CS
M1_nFAULT
M2_CS
M2_nFAULT
M1_PWM_PH
M1_PWM_EN
M2_PWM_PH
M2_PWM_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
RELAY_1
RELAY_2
I2C_SDA
I2C_SCL
IO_INT_N
File: MCU.kicad_sch
DC_Motor_1
M1_PWM_PH
M1_PWM_EN
+3V3_DRV_EN
CSD
nFAULTD
M1_CS
M1_nFAULT
File: Motor_Channel.kicad_sch
Steppers_and_Relays
+3V3_DRV_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
STEP1_EN
STEP2_EN
RELAY_1
RELAY_2
File: Steppers_Relays.kicad_sch
Industrial_IO
I2C_SDA
I2C_SCL
+3V3_DRV_EN
IO_INT_ND
STEP1_END
STEP2_END
File: Industrial_IO.kicad_sch
DC_Motor_2
M2_PWM_PH
M2_PWM_EN
+3V3_DRV_EN
CSD
nFAULTD
M2_CS
M2_nFAULT
File: Motor_Channel.kicad_sch
Sheet: /
File: abx-motion.kicad_sch
Title: ABX-Motion
Size: A4
Date:
Rev: 1.1b
KiCad E.D.A. 10.0.7
Id: 1/7

SHEET 1: BATTERY INPUT, REVERSE POLARITY PROTECTION, 5V BUCK & 3.3V LDO

D2: 33V standoff; 40V max at specified 8/20us pulse. No continuous braking-energy rating.

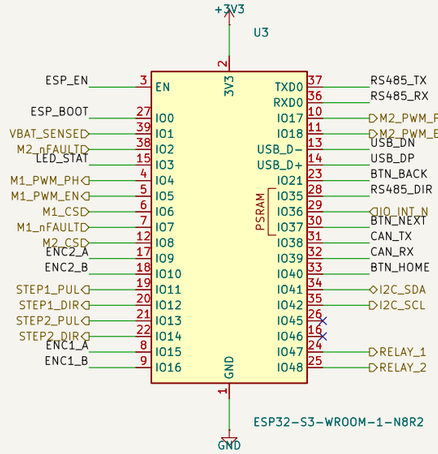


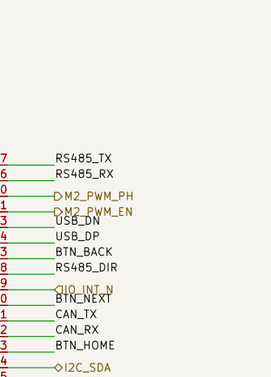
Sheet: /Power_Supply/
File: Power.kicad_sch

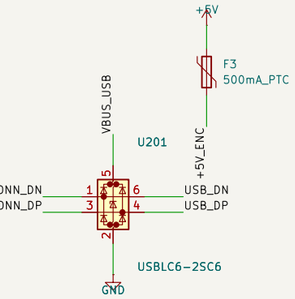
Title: ABX-Motion

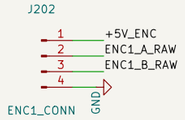
Size: A3	Date:	**Rev: 1.1b**
KiCad E.D.A. 10.0.7		Id: 2/7

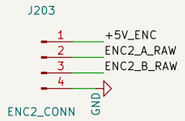
SHEET 2: ESP32-S3-WROOM-1, USB-C, 74LVC14A ENCODERS, CAN, RS-485, OLED & UI BUTTONS

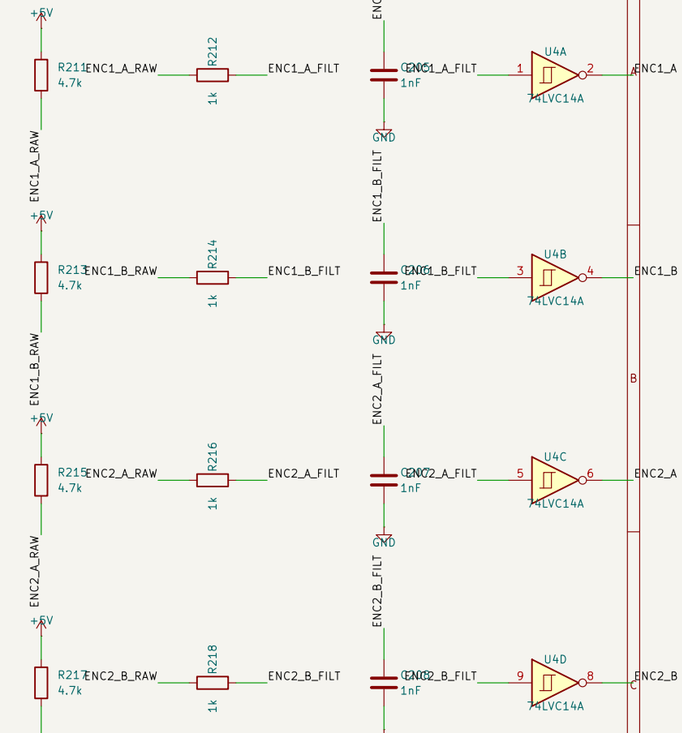


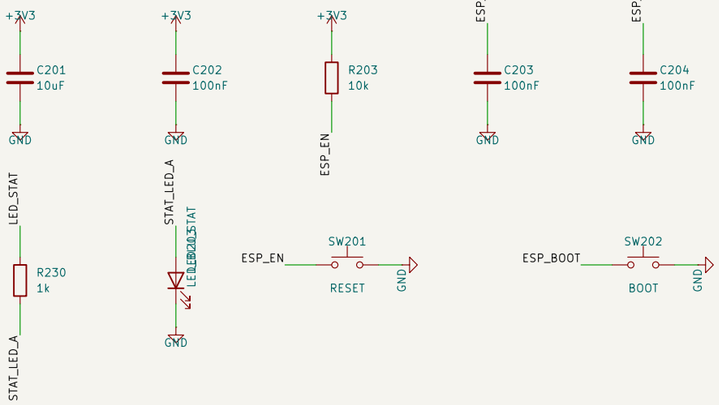


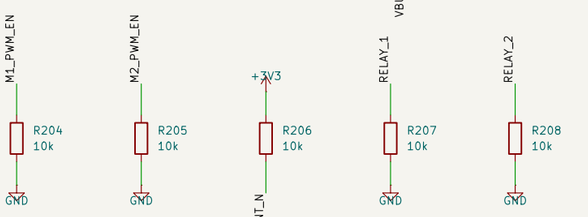


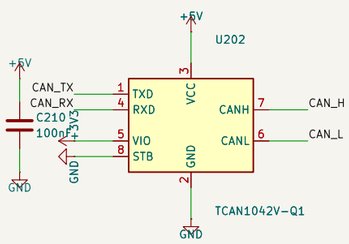


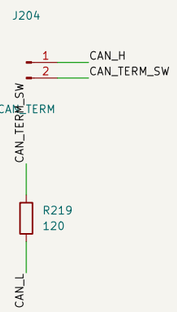


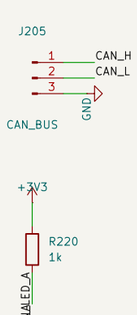


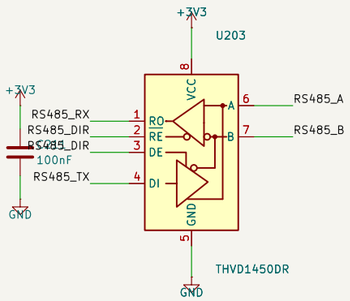


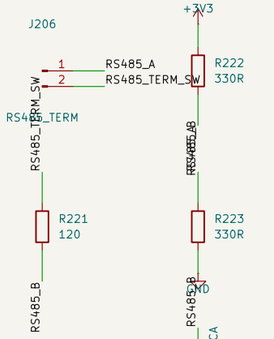


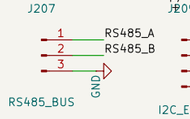


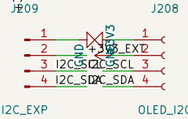




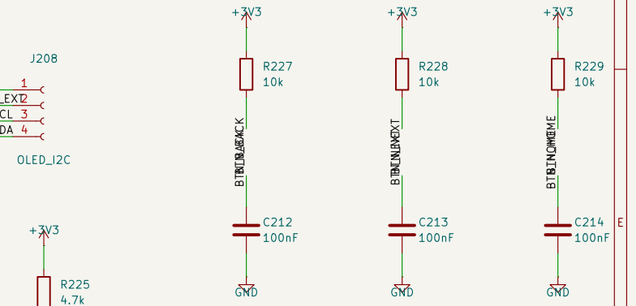


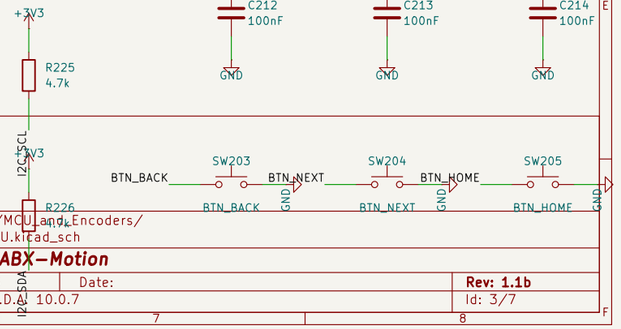










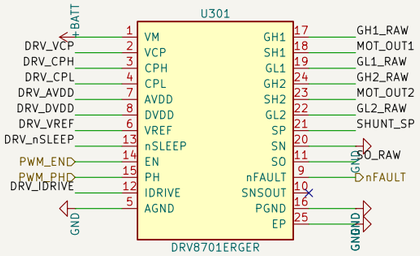


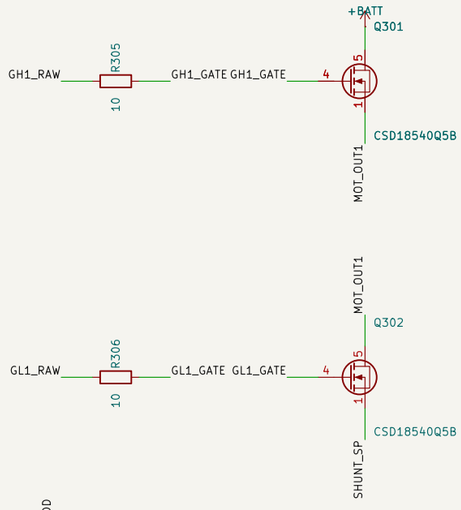
Sheet: /MCU and Encoders/
File: MCU.kicad_sch

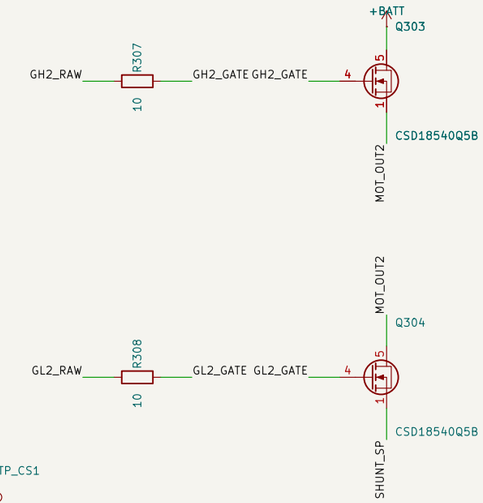
Title: ABX-Motion

Size: A3	Date:	**Rev: 1.1b**
KiCad E.D.A. 10.0.7		Id: 3/7

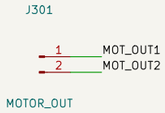
SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)











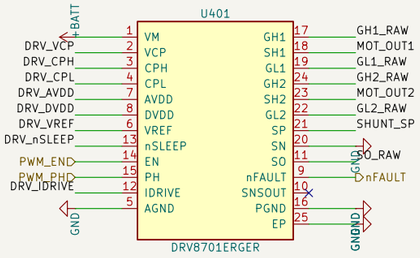


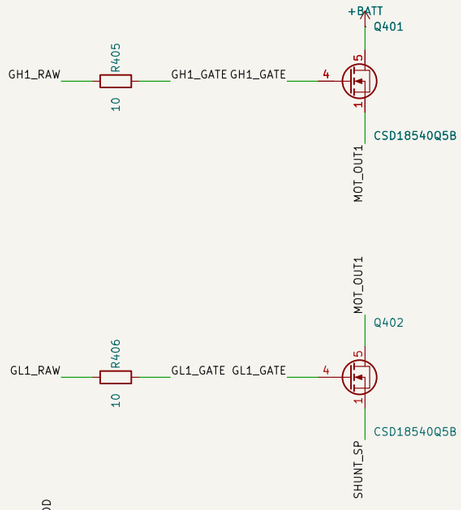


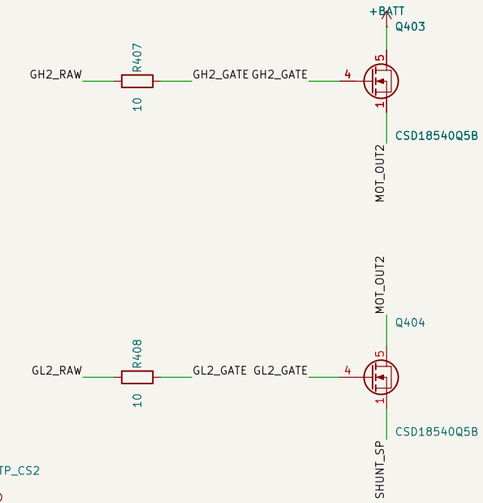
ADC protection: R304/R404 + R313/R413 divide SO by 2; rescale firmware.

Sheet: /DC_Motor_1/		
File: Motor_Channel.kicad_sch		
Title: ABX-Motion		
Size: A3	Date:	**Rev: 1.1b**
KiCad E.D.A. 10.0.7		Id: 4/7

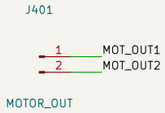
SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)



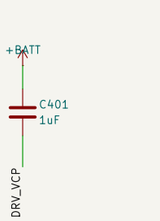




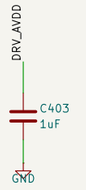


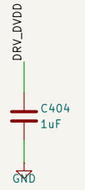




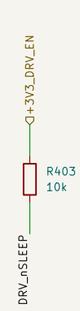


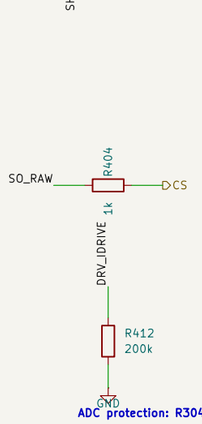




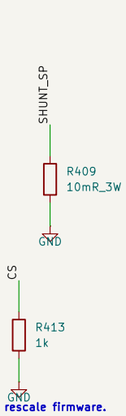


















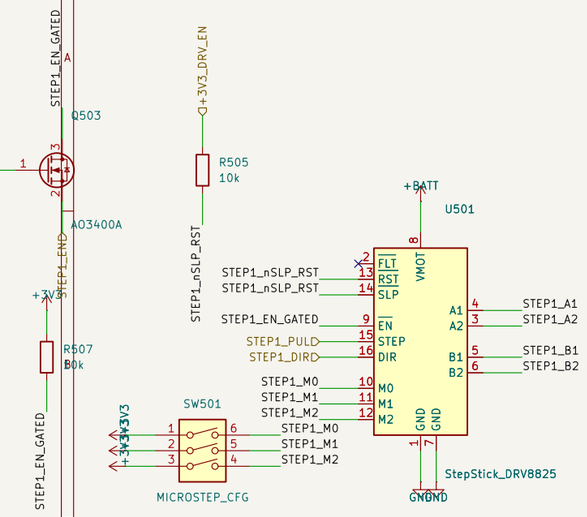


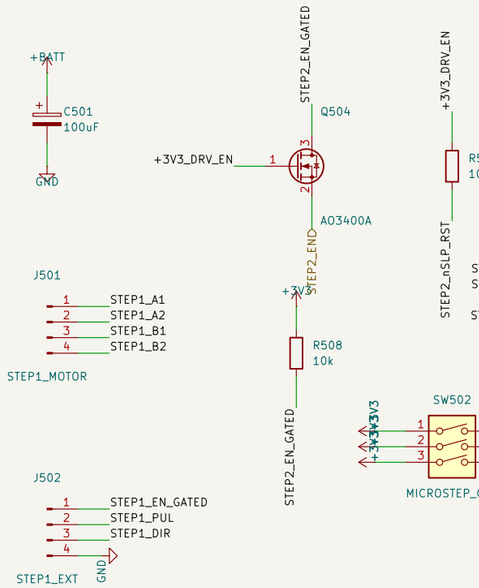


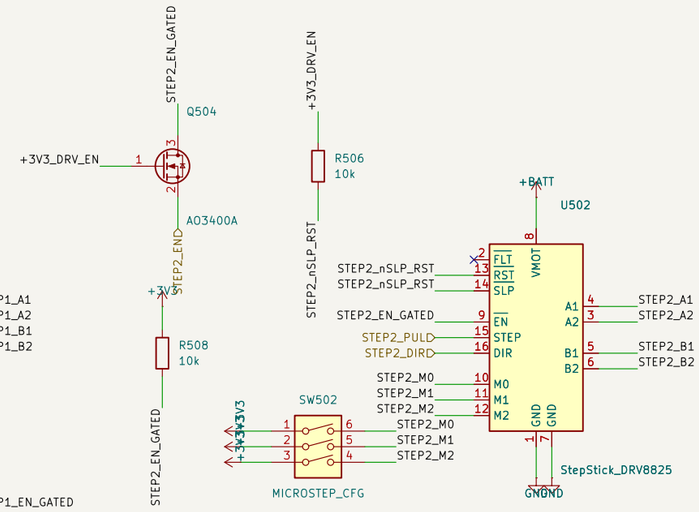
ADC protection: R304/R404 + R313/R413 divide SO by 2; rescale firmware.

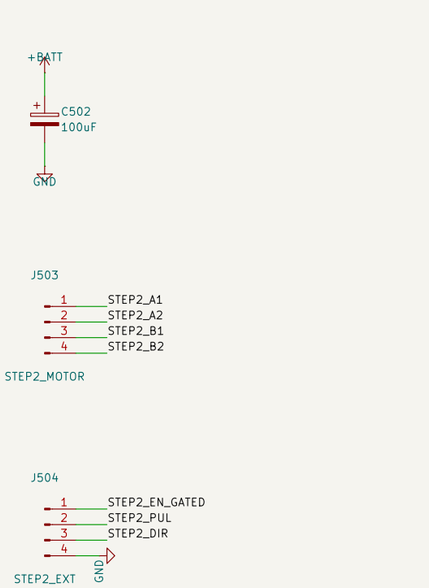
Sheet: /DC_Motor_2/		
File: Motor_Channel.kicad_sch		
Title: ABX-Motion		
Size: A3	Date:	**Rev: 1.1b**
KiCad E.D.A. 10.0.7		Id: 5/7

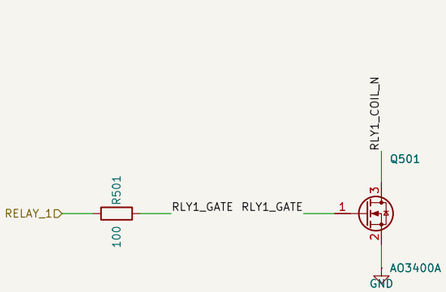
SHEET 4: DUAL STEPSTICK STEPPER CHANNELS & DUAL 5V OMRON G5LE RELAY OUTPUTS

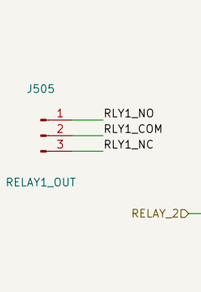


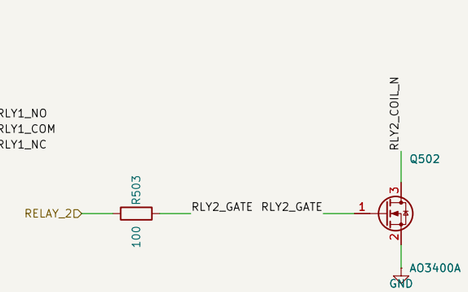


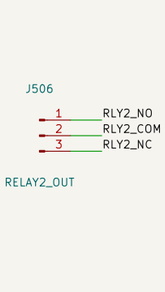












Sheet: /Steppers_and_Relays/		
File: Steppers_Relays.kicad_sch		
Title: ABX—Motion		
Size: A3	Date:	**Rev: 1.1b**
KiCad E.D.A. 10.0.7		Id: 6/7

SHEET 5: 6x 3-PIN JST-PH INDUSTRIAL I/O (8x LIMIT/GPIO + 2x 5V/500mA DC OUTPUTS) & PCA9555PW EXPANDER

J601
M1_MIN_RAW
M1_MAX_RAW
GND
B3B-PH-K-S

J602
M2_MIN_RAW
M2_MAX_RAW
GND
B3B-PH-K-S

J603
S1_MIN_RAW
S1_MAX_RAW
GND
B3B-PH-K-S

J604
S2_MIN_RAW
S2_MAX_RAW
GND
B3B-PH-K-S

+3V3
R601 4.7k
R602 4.7k
R603 4.7k
R604 4.7k
R605 4.7k
R606 4.7k
R607 4.7k
R608 4.7k
M1_MIN_RAW
M1_MAX_RAW
M2_MIN_RAW
M2_MAX_RAW
S1_MIN_RAW
S1_MAX_RAW
S2_MIN_RAW
S2_MAX_RAW

R609 1k
R610 1k
R611 1k
R612 1k
R613 1k
R614 1k
R615 1k
R616 1k
M1_MIN_RAW → M1_MIN_IN
M1_MAX_RAW → M1_MAX_IN
M2_MIN_RAW → M2_MIN_IN
M2_MAX_RAW → M2_MAX_IN
S1_MIN_RAW → S1_MIN_IN
S1_MAX_RAW → S1_MAX_IN
S2_MIN_RAW → S2_MIN_IN
S2_MAX_RAW → S2_MAX_IN

C602 10nF
C603 10nF
C604 10nF
C605 10nF
C606 10nF
C607 10nF
C608 10nF
C609 10nF
M1_MIN_IN
M1_MAX_IN
M2_MIN_IN
M2_MAX_IN
S1_MIN_IN
S1_MAX_IN
S2_MIN_IN
S2_MAX_IN
GND

U601
+3V3
C601 100nF
I2C_SCLD
I2C_SDA
IO_INT_N
SCL
SDA
INT
VDD
A0
A1
A2
VSS
IO0_0 M1_MIN_IN
IO0_1 M1_MAX_IN
IO0_2 M2_MIN_IN
IO0_3 M2_MAX_IN
IO0_4 S1_MIN_IN
IO0_5 S1_MAX_IN
IO0_6 S2_MIN_IN
IO0_7 S2_MAX_IN
IO1_0 OUT1_CMD
IO1_1 OUT2_CMD
IO1_2 STEP1_EN
IO1_3 STEP2_EN
IO1_4 +3V3_DRV_EN
IO1_5
IO1_6
IO1_7
PCA9555PW,118

+5V
1206L050/63YR
+5V_OUT

OUT1_CMD
R617 1k
R619 100
OUT1_GATE
Q601
OUT1_SW
AO3400A
B5819W
+5V_OUT
OUT1_SW
J605
+5V_OUT
OUT1_SW
B3B-PH-K-S

OUT2_CMD
R618 1k
R620 100
OUT2_GATE
Q602
OUT2_SW
AO3400A
B5819W
+5V_OUT
OUT2_SW
J606
+5V_OUT
OUT2_SW
B3B-PH-K-S

R621 10k
R622 10k
STEP1_EN
STEP2_EN

Sheet: /Industrial_IO/
File: Industrial_IO.kicad_sch
Title: ABX-Motion
Size: A3 | Date:
KiCad E.D.A. 10.0.7
Rev: 1.1b
Id: 7/7